Department of Mechanical and Production Engineering

Aarhus University

Exercises

Author: Noah Rahbek Bigum Hansen

Student ID: 202405538

Course: Fluid Mechanics — 290211U004

Date: November 14, 2025

Contents

Lecture 2: Basic equations of fluid statics and buoyancy, surface tension 28. August 2025

1.1 Exercise 3.7

Calculate the absolute pressure and gauge pressure in an open tank of crude oil 2,4 m below the liquid surface. If the tank is closed and pressurized to 130 kPa , that are the absolute pressure and gauge pressure at this location?

It is assumed that the oil is totally incompressible and that the gauge pressure is zero at the liquid surface before the tank is pressurized – i.e. $p_0 = 101\,\text{kPa}$. Also the density of the oil is assumed to be $800\,\text{kg/m}^3$. The formula for the pressure at a depth h in a liquid is:

$$p_{a_1} = p_0 + \rho g h = 101\,\text{kPa} + 800\,\text{kg/m}^3 \cdot 9{,}81\,\text{m/s}^2 \cdot 2{,}4\,\text{m} = 120\,\text{kPa}.$$

The gauge pressure is simply this but without the atmospheric pressure added on:

$$p_{g_1} = \rho g h = 800\,\text{kg/m}^3 \cdot 9{,}81\,\text{m/s}^2 \cdot 2{,}4\,\text{m} = 18{,}8\,\text{kPa}.$$

After the tank is pressurized the gauge pressure will stay constant, whereas the absolute pressure will grow with $p_p = 130\,\text{kPa}$ and become $p_{a_2} = 120\,\text{kPa} + 130\,\text{kPa} = 250\,\text{kPa}$ and $p_{g_2} = 18{,}8\,\text{kPa} + 130\,\text{kPa} = 149\,\text{kPa}$.

1.2 Exercise 3.5

A piston is placed on a tank filled with mercury at 20 °C as shown on **Figure 0.1**. A force is applied to the piston and the height of the mercury column rises. Determine the weight of the piston and the applied force.

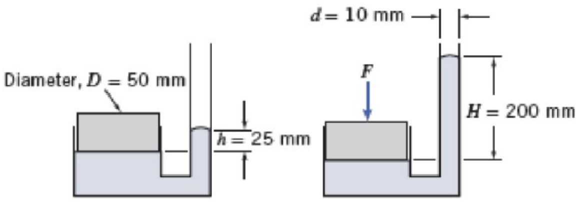


Figure 0.1

For the system to be static the force exerted by the piston on the mercury must be equal to the force exerted by the column. When no force other than the weight of the piston itself is applied a mercury cylinder of height 25 mm and diameter 10 mm is produced. The height rise of this must give rise to a pressure increase given by:

$$\Delta p = \rho g \Delta H.$$

The piston must therefore supply a weight big enough to bring about this pressure. I.e.

$$W = \Delta p A_p = \rho g \Delta H \cdot \pi \cdot r_p^2 = 13{,}5\,\text{g/cm}^3 \cdot 9{,}81\,\text{m/s}^2 \cdot 25\,\text{mm} \cdot \pi \cdot (25\,\text{mm})^2 = 0{,}26\,\text{N}.$$

When the additional force is added the column rises an additional $\Delta H = 175\,\text{mm}$. Therefore we here get:

$$F = \Delta p A_p = 13{,}5\,\text{g/cm}^3 \cdot 9{,}81\,\text{m/s}^2 \cdot 175\,\text{mm} \cdot \pi \cdot (25\,\text{mm})^2 = 45{,}5\,\text{N}.$$

1.3 Exercise 3.6

A 125 mL cube of solid oak is held submerged by a tether as shown on **Figure 0.2**. Calculate the force of the water on the bottom surface of the cube and the tension in the tether.

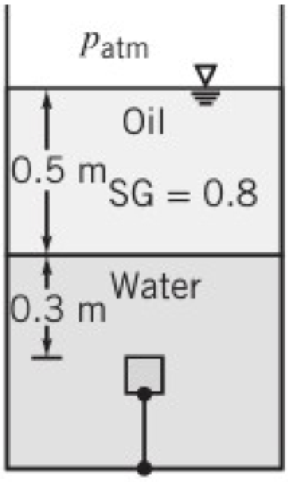


Figure 0.2

First the side length of the cube is found as

$$s = \sqrt[3]{V_{\text{oak}}} = \sqrt[3]{125\,\text{cm}^3} = 5\,\text{cm}^3.$$

Now we can determine its weight as the specific gravity of oak is 0,77 as:

$$W = V_{\text{oak}}\rho \text{SG}_{\text{oak}} g = 125\,\text{mL} \cdot 1000\,\text{kg/m}^3 \cdot 0{,}77 \cdot 9{,}81\,\text{m/s}^2 = 0{,}944\,\text{N}.$$

The force on the bottom surface is equal to the pressure at this depth multiplied by the surface area of the bottom face. Firstly the pressure is found as:

$$\begin{aligned} p_L &= p_{\text{atm}} + \text{SG}_{\text{oil}} \cdot \rho \cdot g \cdot h_{\text{oil}} + \rho g h_L \\ &= p_{\text{atm}} + \rho g \left(\text{SG}_{\text{oil}} \cdot h_{\text{oil}} + h_L\right) \\ &= 101\,\text{kPa} + 1000\,\text{kg/cm}^3 \cdot 9{,}81\,\text{m/s}^2 \left(0{,}8 \cdot 0{,}5\,\text{m} + 0{,}35\,\text{m}\right) \\ &= 109\,\text{kPa}. \end{aligned}$$

Now this can be multiplied by the surface area of the bottom as:

$$F_L = p_L \cdot A = 109\,\text{kPa} \cdot 25\,\text{cm}^2 = 272\,\text{N}.$$

Therefore the force exerted by the fluid on the bottom face is about 272 N.

To find the tension in the tether we realize that the force exerted by the fluid on the bottom face is counteracted by a force exerted by the fluid on the top face. We can find this using the same procedure as above.

$$F_U = \left(p_{\text{atm}} + \rho g \left(\text{SG}_{\text{oil}} \cdot h_{\text{oil}} + h_U\right)\right) \cdot A.$$

We can calculate the resultant force exerted by the liquid on the cube as:

$$\Delta F = F_L - F_U = \rho g \cdot (h_L - h_U) \cdot A = 1000\,\text{kg/m}^3 \cdot 9{,}81\,\text{m/s}^2 \cdot (5\,\text{cm}) \cdot 25\,\text{cm}^2 = 1{,}23\,\text{N}.$$

This force (directed directly upwards) is counteracted by the weight of the cube. I.e.

$$T = \Delta F - W = 1{,}23\,\text{N} - 0{,}944\,\text{N} = 0{,}282\,\text{N}.$$

1.4 Exercise 3.43

Determine the specific weight of the cube when one half is submerged as shown on **Figure 0.3**. Determine the position of the center of the cube relative to the water level when the weight is removed.

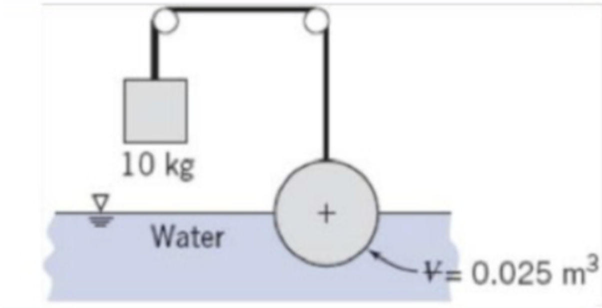


Figure 0.3: Caption

For static equilibrium the sum of the forces in the upwards/downwards direction must be 0. The only forces acting in this direction on the sphere is its weight (downwards) and the tension and the bouyancy forces (upwards). The tension in the cable must be exactly enough to keep the cube steady:

$$T = M \cdot g.$$

The buoyancy force on the sphere is given by Archimedes principle as:

$$F_B = \rho \cdot g \cdot \frac{V}{2}.$$

The weight of the sphere is given by:

$$W = \mathrm{SG} \cdot \rho \cdot g \cdot V.$$

These must, as mentioned, all sum to zero as:

$$Mg + \rho g \frac{V}{2} - \mathrm{SG}\rho g V = 0 \implies \mathrm{SG} = \frac{M}{\rho V} + \frac{1}{2}.$$

Now known values can be substituted in as:

$$\mathrm{SG} = 10\,\mathrm{kg} \cdot \frac{1}{1000\,\mathrm{kg/m^3} \cdot 0{,}025\,\mathrm{m^3}} + \frac{1}{2} = 0{,}9.$$

The specific weight is

$$\gamma = \frac{\text{Weight}}{\text{Volume}} = \frac{\mathrm{SG} \cdot \rho \cdot g \cdot V}{V} = \mathrm{SG}\rho \cdot g = 0{,}9 \cdot 1000\,\mathrm{kg/m^3} \cdot 9{,}81\,\mathrm{m/s^2} = 8830\,\mathrm{N/m^3}.$$

Now to find the equilibrium position when floating we repeat the force balance, but this time with $T = 0$ as

$$F_B - W = 0 \implies W = F_B = \rho g V_{\text{submerged}}.$$

Now we must realize that the submerged part of the sphere will be a spherical cap. The volume of such an object is:

$$V_{\text{submerged}} = \frac{\pi \cdot h_{\text{submerged}^2}}{3} \cdot \left(3 \cdot R - h_{\text{submerged}}\right).$$

Where R is the radius of the sphere. The radius of the sphere can be found as:

$$
\begin{aligned}
V &= \frac{4}{3}\pi R^3 \\
R &= \sqrt[3]{\frac{3V}{4\pi}} \\
&= \sqrt[3]{\frac{3}{4\pi}\cdot 0{,}025\,\mathrm{m}^3} \\
&= 0{,}181\,\mathrm{m}.
\end{aligned}
$$

Therefore:

$$
\begin{aligned}
W &= \rho g V_{\text{submerged}} \\
\mathrm{SG}\cdot\rho g V &= \rho g \frac{\pi h_{\text{submerged}^2}}{3}\cdot\left(3\cdot R - h_{\text{submerged}}\right) \\
\mathrm{SG}\cdot V &= \frac{\pi h_{\text{submerged}^2}}{3}\left(3R - h_{\text{submerged}}\right) \\
\frac{3\mathrm{SG}\cdot V}{\pi} &= h^2_{\text{submerged}}\left(3R - h_{\text{submerged}}\right) \\
h^2_{\text{submerged}}\left(3\cdot 0{,}181\,\mathrm{m} - h_{\text{submerged}}\right) &= \frac{3\cdot 0{,}9\cdot 0{,}025\,\mathrm{m}^3}{\pi} \\
h^2_{\text{submerged}}\left(0{,}544\,\mathrm{m} - h_{\text{submerged}}\right) &= 0{,}0215\,\mathrm{m}^3 \\
\implies h_{\text{submerged}} &\in \{-0{,}173\,\mathrm{m}, 0{,}292\,\mathrm{m}, 0{,}425\,\mathrm{m}\}.
\end{aligned}
$$

Here the negative solution can be rejected as it is unphysical. The largest solution can also be rejected as $0{,}425\,\mathrm{m} > 2\cdot R = 0{,}362\,\mathrm{m}$ and this would require the sphere to be floating beneath the water surface, which is also unphysical. Therefore the sphere must be floating at a height of $h = 0{,}292\,\mathrm{m}$.

Lecture 3: Control volume analysis I: Basic laws and mass conservation

1. September 2025

2.1 Exercise 4.5

A 0,3 m by 0,5 m rectangular air duct carries a flow of $0{,}45\,\mathrm{m}^3/\mathrm{s}$ at a density of $2\,\mathrm{kg}/\mathrm{m}^3$, Calculate the mean velocity in the duct. If the duct tapers to 0,15 m by 0,5 m size, determine the mean velocity in this section if the density is $1{,}5\,\mathrm{kg}/\mathrm{m}^3$ in this section.

In the initial state $0{,}45\,\mathrm{m}^3/\mathrm{s}$ of air is passing by an area 0,3 m by 0,5 m. This area is:

$$A = 0{,}3\,\mathrm{m}\cdot 0{,}5\,\mathrm{m} = 0{,}15\,\mathrm{m}^2.$$

Now if we divide the flow by the area through which it passes we will obtain the velocity of the air as:

$$V_1 = \frac{0{,}45\,\frac{\mathrm{m}^3}{\mathrm{s}}}{0{,}15\,\mathrm{m}^2} = 3\,\frac{\mathrm{m}}{\mathrm{s}}.$$

Therefore the air will initially be moving at a velocity of $V_1 = 3\,\frac{\mathrm{m}}{\mathrm{s}}$.

As mass if a conserved property the mass flow in the initial section must equal the mass flow of the tapered section. Using the continuity equation we can calculate the mass flow rate initially as:

$$\dot m_1 = \rho_1 V_1 A_1 = 2\,\frac{\mathrm{kg}}{\mathrm{m}^3}\cdot 3\,\frac{\mathrm{m}}{\mathrm{s}}\cdot 0{,}15\,\mathrm{m}^2 = 0{,}9\,\frac{\mathrm{kg}}{\mathrm{s}}.$$

This must equal the mass flow rate after the taper, thus:

$$\begin{aligned}
\dot{m}_1 &= \dot{m}_2 \\
\dot{m}_1 &= \rho_2 V_2 A_2 \\
V_2 &= \frac{\dot{m}_1}{\rho_2 A_2} \\
V_2 &= \frac{0{,}9\,\frac{\text{kg}}{\text{s}}}{1{,}5\,\frac{\text{kg}}{\text{m}^3}\cdot 0{,}15\,\text{m}\cdot 0{,}5\,\text{m}} \\
V_2 &= 8\,\frac{\text{m}}{\text{s}}.
\end{aligned}$$

Therefore the mean velocity of the tapered section is 8 m/s.

2.2 Exercise 4.9

A pipeline 0,3 m in diameter divides at a Y into two branches 200 mm and 150 mm in diameter. If the flow rate in the main line is 0,3 m^3/s and the mean velocity in the 200 mm pipe is 2,5 m/s, determine the flow rate in the 150 mm pipe.

We assume that the water flow can be modelled as steady and incompressible through a fixed control volume. In this case the continuity equation simplifies to:

$$\int_{\text{CS}} \mathbf{V}\cdot \mathrm{d}\mathbf{A} = 0$$

or

$$V_1 A_1 = V_2 A_2 + V_3 A_3.$$

Using the flow rates $Q_1 = V_1 A_1$, $Q_2 = V_2 A_2$, and $Q_3 = V_3 A_3$ we can write:

$$Q_1 = Q_2 + Q_3.$$

Let the 150 mm pipe be number 3 and we can then isolate the flow rate we wish to find Q_3 in this as:

$$Q_3 = Q_1 - Q_2.$$

We can now find the flow rate of Q_2 as

$$Q_2 = V_2 A_2 = 2{,}5\,\frac{\text{m}}{\text{s}}\cdot \pi \cdot (100\,\text{mm})^2 = 0{,}0785\,\frac{\text{m}^3}{\text{s}}.$$

And thus the flow rate in the 150 mm pipe Q_3 can be found as:

$$Q_3 = 0{,}3\,\frac{\text{m}^3}{\text{s}} - 0{,}0785\,\frac{\text{m}^3}{\text{s}} = 0{,}221\,\frac{\text{m}^3}{\text{s}}.$$

2.3 Exercise 4.12

A cylindrical tank, of diameter $D = 150$ mm, drains through an opening, $d = 5$ mm, in the bottom of the tank. The speed of the liquid leaving the tank is approximately $V = \sqrt{2gy}$, where y is the height from the tank bottom to the free surface. If the tank is initially filled with water to $y_0 = 0{,}4$ m, determine the water depths at $t = 60$ s, $t = 120$ s, and $t = 180$ s. Plot $y(m)$ versus t for the first 180 s.

We assume uniform incompressible flow. The continuity equation states:

$$\frac{\partial}{\partial t}\int_{\mathrm{CV}}\rho\,\mathrm{d}V+\int_{\mathrm{CS}}\rho\mathbf{V}\cdot\mathrm{d}\mathbf{A}=0.$$

In other words, the rate increase of mass in the control volume plus the mass outflow from the control volume equals zero. If we treat the tank as out control volume we get:

$$\begin{aligned}\frac{\partial}{\partial t}\int_0^y\rho\cdot A_{\text{tank}}\,\mathrm{d}y+\rho\cdot V\cdot A_{\text{opening}}&=0\\ \rho\cdot\pi\cdot R^2\cdot\frac{\mathrm{d}y}{\mathrm{d}t}+\rho\cdot\pi\cdot r^2\cdot V&=0.\end{aligned}$$

We can now use $V=\sqrt{2gy}$ and get:

$$\begin{aligned}\rho\cdot\pi\cdot R^2\cdot\frac{\mathrm{d}y}{\mathrm{d}t}+\rho\cdot\pi\cdot r^2\cdot\sqrt{2gy}&=0\\ R^2\frac{\mathrm{d}y}{\mathrm{d}t}+r^2\cdot\sqrt{2gy}&=0\\ \frac{\mathrm{d}y}{\mathrm{d}t}&=-\frac{r^2}{R^2}\cdot\sqrt{2gy}.\end{aligned}$$

Separating variables we get:

$$\frac{1}{y^{\frac{1}{2}}}\,\mathrm{d}y=-\frac{r^2}{R^2}\cdot\sqrt{2g}\,\mathrm{d}t.$$

Integrating we get:

$$\begin{aligned}2\left(y^{\frac{1}{2}}-y_0^{\frac{1}{2}}\right)&=-\frac{r^2}{R^2}\sqrt{2g}\,t\\ y^{\frac{1}{2}}&=-\frac{r^2}{R^2}\sqrt{\frac{g}{2}}\,t+y_0^{\frac{1}{2}}\\ y^{\frac{1}{2}}&=y_0^{\frac{1}{2}}\left(1-\frac{r^2}{R^2}\sqrt{\frac{g}{2y_0}}\,t\right)\\ y&=y_0\left(1-\left(\frac{r}{R}\right)^2\sqrt{\frac{g}{2y_0}}\,t\right)^2.\end{aligned}$$

We have that $y_0=0{,}4\,\mathrm{m}$ and $\frac{r}{R}=\frac{2{,}5\,\mathrm{mm}}{75\,\mathrm{mm}}=\frac{1}{30}$. At $t=60\,\mathrm{s}$ we therefore get:

$$y_{60\,\mathrm{s}}=0{,}4\,\mathrm{m}\cdot\left(1-\left(\frac{1}{30}\right)^2\sqrt{\frac{9{,}81\,\frac{\mathrm{m}}{\mathrm{s}^2}}{2\cdot0{,}4\,\mathrm{m}}}\cdot60\,\mathrm{s}\right)^2=0{,}235\,\mathrm{m}.$$

At $t=120\,\mathrm{s}$ we get

$$y_{120\,\mathrm{s}}=0{,}4\,\mathrm{m}\cdot\left(1-\left(\frac{1}{30}\right)^2\sqrt{\frac{9{,}81\,\frac{\mathrm{m}}{\mathrm{s}^2}}{2\cdot0{,}4\,\mathrm{m}}}\cdot120\,\mathrm{s}\right)^2=0{,}114\,\mathrm{m}.$$

And at $t=180\,\mathrm{s}$ we get:

$$y_{180\,\mathrm{s}}=0{,}4\,\mathrm{m}\cdot\left(1-\left(\frac{1}{30}\right)^2\sqrt{\frac{9{,}81\,\frac{\mathrm{m}}{\mathrm{s}^2}}{2\cdot0{,}4\,\mathrm{m}}}\cdot180\,\mathrm{s}\right)^2=0{,}036\,\mathrm{m}.$$

$y(m)$ versus t has been plotted in GeoGebra on Figure 0.4

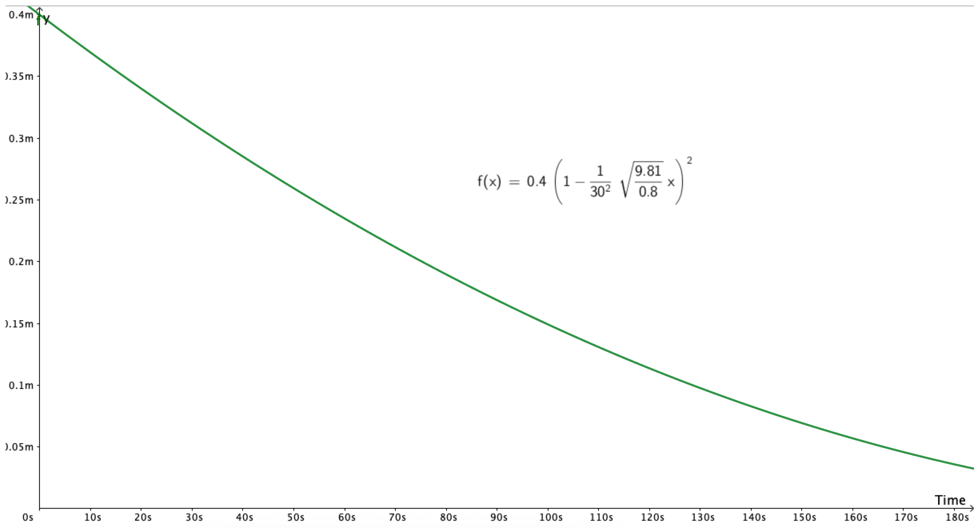
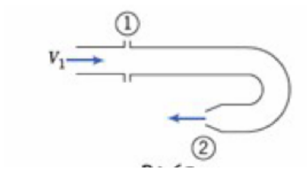


Figure 0.4: Plot of $y(m)$ versus t.

Lecture 4: Control volume analysis II: Momentum & energy equation 3. September 2025

3.1 Exercise 4.17

Water is flowing steadily through the 180° elbow shown. At the inlet to the elbow the gage pressure is 103 kPa. The water discharges to atmospheric pressure. Assume properties are uniform over the inlet and outlet areas: $A_1 = 2500\,\text{mm}^2$, $A_2 = 650\,\text{mm}^2$, and $V_1 = 3\,\frac{\text{m}}{\text{s}}$. Find the horizontal component of force required to hold the elbow in place.



The momentum equation in the x-direction is:

$$R_x + F_{S_x} + F_{B_x} = \frac{\partial}{\partial t}\int_{\text{CV}} u\rho\,\text{d}V + \int_{\text{CS}} u\rho\mathbf{V}\cdot\text{d}\mathbf{A}.$$

As we are seeking the horizontal force on the elbow we can neglect the gravitational force (i.e. $F_{B_x} = 0$). The pressure force is simply:

$$F_{S_x} = p_1 A_1.$$

As we assume steady flow we simply get:

$$\begin{aligned} R_x + p_1 A_1 &= V_1\left(-\rho V_1 A_1\right) - V_2\left(\rho V_2 A_2\right) \\ R_x &= -p_1 A_1 - \rho\left(V_1^2 A_1 + V_2^2 A_2\right). \end{aligned}$$

From the continuity equation we can calculate the outlet velocity V_2 as:

$$\begin{aligned} 0 &= -\rho V_1 A_1 + \rho V_2 A_2 \\ V_2 &= \frac{V_1 A_1}{A_2} \\ V_2 &= 3\,\frac{\mathrm{m}}{\mathrm{s}} \cdot \frac{2500\,\mathrm{mm}^2}{650\,\mathrm{mm}^2} \\ &= 11{,}5\,\frac{\mathrm{m}}{\mathrm{s}}. \end{aligned}$$

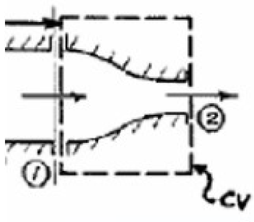
Now all parameters in the equation for R_x are known and it can be found as:

$$\begin{aligned} R_x &= -103\,\mathrm{kPa} \cdot 2500\,\mathrm{mm}^2 - 1000\,\frac{\mathrm{kg}}{\mathrm{m}^3}\left(\left(3\,\frac{\mathrm{m}}{\mathrm{s}}\right)^2 \cdot 2500\,\mathrm{mm}^2 + \left(11{,}5\,\frac{\mathrm{m}}{\mathrm{s}}\right)^2 \cdot 650\,\mathrm{mm}^2\right) \\ &= -367\,\mathrm{N}. \end{aligned}$$

Therefore a force of 367 N directed against the wall is needed to hold the elbow in place.

3.2 Exercise 4.16

Water flows steadily through a fire hose and nozzle. The hose is 35 mm in diameter and the nozzle tip is 25 mm in diameter; water gage pressure in the hose is 510 kPa, and the stream leaving the nozzle is uniform. The exit speed and pressure are 32 m/s and atmospheric, respectively. Find the force transmitted by the coupling between the nozzle and hose. Indicate whether the coupling is in tension or compression.



Here the same formula as above (momentum in a control volume) can be used as:

$$R_x + F_{S_x} + F_{B_x} = \frac{\partial}{\partial t}\int_{\mathrm{CV}} u\rho\,\mathrm{d}V + \int_{\mathrm{CS}} u\rho\mathbf{V}\cdot\mathrm{d}\mathbf{A}.$$

With the same assumption of steady flow as above we get:

$$R_x + p_1 A_1 = V_2\left(\rho V_2 A_2\right) - V_1\left(\rho V_1 A_1\right).$$

For incompressible steady flow we have:

$$0 = -\rho_1 V_1 A_1 + \rho_2 V_2 A_2 \implies V_1 = V_2\frac{A_2}{A_1} = V_2\left(\frac{D_2}{D_1}\right)^2 = 32\,\frac{\mathrm{m}}{\mathrm{s}} \cdot \left(\frac{25\,\mathrm{mm}}{35\,\mathrm{mm}}\right)^2 = 16{,}3\,\frac{\mathrm{m}}{\mathrm{s}}.$$

The condition from above can also be used to rewrite the expression from before as:

$$\begin{aligned} R_x + p_1 A_1 &= V_2\left(\rho V_2 A_2\right) - V_1\left(\rho V_1 A_1\right) \\ R_x &= V_2\left(\rho V_2 A_2\right) - V_1\left(\rho V_1 A_1\right) - p_1 A_1 \\ &= V_2\left(\rho V_2 A_2\right) - V_1\left(\rho V_2 A_2\right) - p_1 A_1 \\ &= \left(\rho V_2 A_2\right)\left(V_2 - V_1\right) - p_1 A_1. \end{aligned}$$

Plugging in all the known values we get:

$$R_x = \left(1000\,\frac{\text{kg}}{\text{m}^3}\cdot 32\,\frac{\text{m}}{\text{s}}\cdot\pi\cdot(12{,}5\,\text{mm})^2\right)\left(32\,\frac{\text{m}}{\text{s}} - 16{,}3\,\frac{\text{m}}{\text{s}}\right) - 510\,\text{kPa}\cdot\pi\cdot(17{,}5\,\text{mm})^2 = -245\,\text{N}.$$

Therefore a force of $R_x = -245\,\text{N}$ would be applied to the nozzle. The sign tells us that the force acts against the control volume to the left meaning the nozzle must itself be in tension.

3.3 Exercise 4.14

The radial-flow turbine in an automotive turbocharger has a diameter of 60 mm and rotates at 100 000 RPM. An air flow of 0,016 kg/s enters the turbine wheel at 60 m/s and an angle of 25° as shown in Figure 0.5. The air flow leaving has negligible angular momentum. Determine the power (W and hp) the turbine produces.

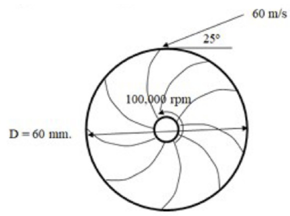


Figure 0.5

In this exercise the angular-momentum principle for an inertial reference system will be used. It is:

$$\mathbf{r}\times\mathbf{F}_s + \int_{\text{CV}} \mathbf{r}\times\mathbf{g}\rho\,\mathrm{d}V + \mathbf{T}_{\text{shaft}} = \frac{\partial}{\partial t}\int_{\text{CV}} \mathbf{r}\times\mathbf{V}\rho\,\mathrm{d}V + \int_{\text{CS}} \mathbf{r}\times\mathbf{V}\rho\mathbf{V}\cdot\mathrm{d}\mathbf{A}.$$

As we assume that there is no body or surface forces acting on the system and that the flow is steady this reduces to:

$$\mathbf{T}_{\text{shaft}} = \int_{\text{CS}} \mathbf{r}\times\mathbf{V}\rho\mathbf{V}\cdot\mathrm{d}\mathbf{A}.$$

In scalar form this becomes

$$T_{\text{shaft}} = r_o V_{\tan}\rho Q = r_0 V_{\tan}\dot{m}.$$

The tangential velocity is the component of the velocity tangent to the impeller as

$$V_{\tan} = 60\,\frac{\text{m}}{\text{s}}\cdot\cos 25^\circ = 54{,}4\,\frac{\text{m}}{\text{s}}.$$

And therefore the torque produced by the rotor is:

$$T_{\text{shaft}} = r_o V_{\tan}\dot{m} = \frac{60\,\text{mm}}{2}\cdot 54{,}4\,\frac{\text{m}}{\text{s}}\cdot 0{,}016\,\frac{\text{kg}}{\text{s}} = 0{,}0261\,\text{N}\,\text{m}.$$

The power produced is given as:

$$\dot{W} = T_{\text{shaft}}\omega$$

where ω is the rotational frequency. The speed of 1600 RPM is therefore converted to frequency as:

$$\omega = 100\,000\,\text{RPM}\cdot 2\pi\text{rev}^{-1}\cdot\frac{\text{min}}{60\,\text{s}} = 10\,500\,\text{s}^{-1}.$$

Therefore the power is:

$$\dot{W} = T_{\text{shaft}}\cdot\omega = 0{,}0261\,\text{N}\,\text{m}\cdot 10\,500\,\text{s}^{-1} = 273\,\text{W} = 0{,}367\,\text{hp}.$$

Lecture 5: Differential Analysis I: Mass conservation

8. September 2025

4.1 Exercise 5.1

Determine which of the following velocity distributions are possibly three-dimensional incompressible flows.

(a)

$$u = 2y^2 + 2xz: \quad v = -2xy + 6x^2yz; \quad w = 3x^2z^2 + x^3y^4.$$

For an incompressible flow the continuity equation simplifies to

$$\nabla \cdot \mathbf{V} = 0 \implies \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0.$$

If we actually carry out this we get:

$$\frac{\partial 2y^2 + 2xz}{\partial x} + \frac{\partial -2yz + 6x^2yz}{\partial y} + \frac{\partial 3x^2z^2 + x^3y^4}{\partial z} = 2z - 2z + 6x^2z + 6x^2z$$
$$12x^2z \neq 0.$$

Hence the equality does not hold and a flow with the given velocity field must therefore at the very least be compressible.

(b)

$$u = xyzt; \quad v = -xyzt^2; \quad w = z^2\left(xt^2 - yt\right).$$

The same procedure is followed here to obtain

$$\frac{\partial xyzt}{\partial x} + \frac{\partial -xyzt^2}{\partial y} + \frac{\partial z^2\left(xt^2 - yt\right)}{\partial z} = yzt - xzt^2 + 2z\left(xt^2 - yt\right) \neq 0.$$

Hence the fluid most once again be compressible.

(c)

$$u = x^2 + 2y + z^2; \quad v = x - 2y + z; w = -2xz + y^2 + 2z.$$

Once again the same procedure is followed:

$$\frac{\partial x^2 + 2y + 2z^2}{\partial x} + \frac{\partial x - 2y + z}{\partial y} + \frac{\partial -2xz + y^2 + 2z}{\partial z} = 2x - 2 - 2x + 2 = 0.$$

Hence, in this case, the fluid is incompressible.

4.2 Exercise 5.3

The x component of velocity in a steady, incompressible flow field in the xy plane is $u = \frac{A}{x}$, where $A = 2\,\frac{\mathrm{m}^2}{\mathrm{s}}$, and x is measured in meters. Find the simplest y component of velocity for this flow field.

The continuity equation for a two-dimensional incompressible steady flow may be written as:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \implies \frac{\partial u}{\partial x} = -\frac{\partial v}{\partial y}.$$

As we have $u = \frac{A}{x}$ the partial of u with respect to x is:

$$\frac{\partial u}{\partial x} = -\frac{A}{x^2}.$$

Therefore we get:

$$\begin{aligned}\frac{\partial u}{\partial x} &= -\frac{\partial v}{\partial y}\\ -\frac{A}{x^2} &= -\frac{\partial v}{\partial y}\\ \frac{\partial v}{\partial y} &= \frac{A}{x^2}.\end{aligned}$$

We now integrate this:

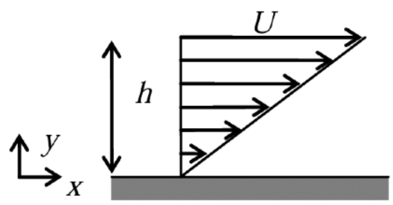
$$v = \int \frac{A}{x^2}\,\mathrm{d}y = \frac{A \cdot y}{x^2} + f(x).$$

The simplest version of this is for $f(x) = 0$ so:

$$v = \frac{A \cdot y}{x^2}.$$

4.3 Exercise 5.12

The velocity in a parallel one-dimensional flow in the positive x direction varies linearly from zero at $y = 0$ to $30\,\mathrm{m/s}$ at $y = 1{,}5\,\mathrm{m}$. Determine the stream function. Calculate the volume flow rate and the value of y at one-half of the total flow.



The flow rate in the x direction can be described by:

$$u = U\left(\frac{y}{h}\right).$$

And the flow in the y direction by:

$$v = 0.$$

We start by checking if the flow is incompressible:

$$\begin{aligned}\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} &= 0\\ \frac{\partial U\left(\frac{y}{h}\right)}{\partial x} + \frac{\partial 0}{\partial y} &= 0.\end{aligned}$$

Hence the flow is incompressible. The stream function ψ for an incompressible flow is defined by the two equations:

$$u = \frac{\partial \psi}{\partial y}; \qquad v = -\frac{\partial \psi}{\partial x}.$$

We integrate the expression for u with respect to y to find the stream function:

$$\begin{aligned}\psi(x,y) &= \int U\frac{y}{h}\,\mathrm{d}y \\ &= \frac{Uy^2}{2h} + f(x).\end{aligned}$$

And we can also integrate the expression for v with respect to x to get:

$$\begin{aligned}\psi(x,y) &= -\int 0\,\mathrm{d}x \\ &= g(y).\end{aligned}$$

As these two functions must be the same. Therefore $f(x) = 0$ and $g(y) = \frac{U2y^2}{2h}$. The stream function therefore becomes:

$$\psi(x,y) = \frac{Uy^2}{2h}.$$

The entire flow rate can be found by integrating the velocity over the entire range. In this case:

$$Q = \int_0^h u\,\mathrm{d}y = \frac{U}{h}\int_0^h y\,\mathrm{d}y = \frac{Uh}{2}.$$

Now to find the y-value at half the flow rate we use the same formula however we plug $\frac{Q}{2}$ in at the place of the flow rate and set $h = h_{\text{half}}$ as our variable as:

$$\begin{aligned}\frac{Q}{2} &= \int_0^{h_{\text{half}}} u\,\mathrm{d}y \\ &= \frac{U}{h}\int_0^{h_{\text{half}}} y\,\mathrm{d}y \\ &= \frac{U}{h}\cdot\frac{h_{\text{half}}^2}{2} \\ \frac{Q}{2} &= \frac{1}{2}\cdot\left(\frac{Uh}{2}\right) \\ &= \frac{U\cdot h}{4} \\ \implies \frac{U}{h}\cdot\frac{h_{\text{half}}^2}{2} &= \frac{U\cdot h}{4} \\ h_{\text{half}}^2 &= \frac{h^2}{2} \\ &= \frac{(1{,}5\,\mathrm{m})^2}{2} \\ &= 1{,}06\,\mathrm{m}.\end{aligned}$$

Lecture 6: Differential Analysis II: Kinematics & Navier-Stokes equations 10. September 2025

5.1 Exercise 5.15

A 4 m diameter tank is filled with water and then rotated at a rate of $\omega = 2\pi\left(1 - e^{-t}\right)\mathrm{s}^{-1}$. At the tank walls, viscosity prevents relative motion between the fluid and the wall. Determine the speed and acceleration of the fluid particles next to the tank walls as a function of time.

We assume steady and incompressible flow. Due to viscosity the boundary condition at the tank wall requires that the velocity of the fluid particles right next to the walls must therefore be the same as that of the wall, i.e.

$$V = \omega r = \frac{1}{2}\omega d = \frac{1}{2}\cdot 2\pi\left(1-e^{-t}\right)\mathrm{s}^{-1}\cdot 4\,\mathrm{m} = 4\pi\left(1-e^{-t}\right)\frac{\mathrm{m}}{\mathrm{s}}.$$

The tangential acceleration is thus:

$$a_t = \frac{\mathrm{d}V}{\mathrm{d}t} = 4\pi e^{-t}\frac{\mathrm{m}}{\mathrm{s}^2}.$$

And the normal acceleration is:

$$a_n = -\frac{V^2}{r} = -\frac{16\pi^2\left(1-e^{-t}\right)^2\frac{\mathrm{m}^2}{\mathrm{s}^2}}{2\,\mathrm{m}} = -8\pi^2\left(1-e^{-t}\right)^2\frac{\mathrm{m}}{\mathrm{s}^2}.$$

5.2 Exercise 5.19

A velocity field is given $\mathbf{V} = 2\hat{i} - 4x\hat{j}\,\frac{\mathrm{m}}{\mathrm{s}}$. Determine an equation for the streamline. Calculate the vorticity of the flow.

The given velocity field is:

$$\begin{aligned} u &= 2\,\frac{\mathrm{m}}{\mathrm{s}} \\ v &= -4x\frac{\mathrm{m}}{\mathrm{s}}. \end{aligned}$$

To fund an equation for the streamline we use that the x component of the velocity in terms of the stream function is:

$$u = \frac{\partial\psi}{\partial y} = 2.$$

By integrating with respect to y we get

$$\psi = 2y + f(x).$$

Doing the same for the y component we get

$$\begin{aligned} v = -\frac{\partial\psi}{\partial x} &= 4x \\ \frac{\partial\psi}{\partial x} &= -4x \\ \psi &= -2x^2 + f(y). \end{aligned}$$

Hence:

$$\psi = 2y - 2x^2 + c$$

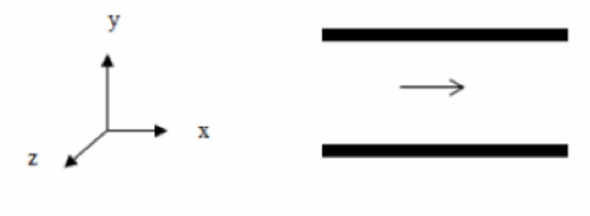
where $c =$ constant.

The vorticity of the flow is calculated as:

$$\boldsymbol{\xi} = \nabla\times\mathbf{V} = \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right)\hat{k} = -4\hat{k}.$$

5.3 Exercise 5.22

Consider a steady, laminar, fully developed incompressible flow between two infinite parallel plates as shown. The flow is due to a pressure gradient applied in the x-direction. Given that $\mathbf{V} \neq \mathbf{V}(z)$, $w = 0$ and that gravity points in the negative y direction, prove that $v = 0$ and that the pressure gradients in the x- and y-directions are constant.



For an incompressible steady flow the continuity equation simplifies to:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0.$$

And the momentum equations for the $x-$, $y-$, and z-directions are

$$\rho\left(u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + w\frac{\partial u}{\partial z}\right) = -\frac{\partial p}{\partial x} + \mu\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}\right)$$
$$\rho\left(u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} + w\frac{\partial v}{\partial z}\right) = -\rho g - \frac{\partial p}{\partial y} + \mu\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2}\right)$$
$$\rho\left(u\frac{\partial w}{\partial x} + v\frac{\partial w}{\partial y} + w\frac{\partial w}{\partial z}\right) = -\frac{\partial p}{\partial z} + \mu\left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2}\right).$$

We have been informed that $w = 0$, $\mathbf{V} \neq \mathbf{V}(z)$ and since the flow is fully developed we also have $\frac{\partial u}{\partial x} = 0$. Plugging this into the continuity equation we get:

$$0 + \frac{\partial v}{\partial y} + 0 = 0$$
$$\implies \frac{\partial v}{\partial y} = 0.$$

Therefore v = constant. As $v = 0$ at the plate (for the continuity equation not to break for viscous flow) $v = 0$ everywhere.

From the momentum equation in the z-direction (since $w = 0$) we get:

$$\frac{\partial p}{\partial z} = 0.$$

From the y-direction equation we get

.

$$\rho\left(u \cdot 0 + 0 \cdot 0 + 0 \cdot 0\right) = -\rho g - \frac{\partial p}{\partial y}$$
$$0 = -\rho g - \frac{\partial p}{\partial y}$$
$$\frac{\partial p}{\partial y} = -\rho g$$
$$\frac{\partial p}{\partial y} = \text{constant}.$$

For the momentum equation in the x-direction we get

$$\rho\left(u\cdot 0+0\cdot\frac{\partial u}{\partial y}+0\cdot 0\right)=-\frac{\partial p}{\partial x}+\mu\left(0+\frac{\partial^2 u}{\partial y^2}+0\right)$$
$$0=-\frac{\partial p}{\partial x}+\mu\left(\frac{\partial^2 u}{\partial y^2}\right)$$
$$\frac{\partial p}{\partial x}=\mu\left(\frac{\partial^2 u}{\partial y^2}\right).$$

This is clearly not a function of x and z however it still could be a function of y. However as

$$\frac{\partial p}{\partial y}=\text{constant.}$$

we know that $\frac{\partial p}{\partial x}$ cannot be a function of y. Hence $\frac{\partial p}{\partial x}=\text{constant}$

Lecture 7: Euler and Bernoulli Equation

15. September 2025

Problem 6.11

An open-circuit wind tunnel draws in air from the atmosphere through a well-contoured nozzle. In the test section, where the flow is straight and nearly uniform, a static pressure tap is drilled into the tunnel wall. A manometer connected to the tap shows that static pressure within the tunnel is 45 mm of water below atmospheric. Assume that the air is incompressible, and at 25 °C, 100 kPa absolute. Calculate the air speed in the wind-tunnel test section.

Assumptions

- Steady, incompressible, inviscid, and uniform flow
- The air behaves as an ideal gas

Derivation

The Bernoulli equation states that

$$\frac{p}{\rho}+\frac{V^2}{2}+gz=\text{constant.}$$

In this case we assume only horizontal flow, hence $gz=0$, furthermore, the air is assumed to be stationary at the entrance, hence $V_1=0$. Therefore

$$\frac{p_1}{\rho}=\frac{p_2}{\rho}+\frac{V_2^2}{2}\implies V_2=\sqrt{2\cdot\frac{p_1-p_2}{\rho}}.$$

The manometer reading corresponds to

$$p_1-p_2=\rho_{\mathrm{H_2O}}gh.$$

Hence,

$$V_1=\sqrt{\frac{2\rho_{\mathrm{H_2O}}gh}{\rho}}.$$

The density of the air can be found using the ideal gas law

$$\rho=\frac{p}{RT}=\frac{100\,\mathrm{kPa}}{287\,\frac{\mathrm{J}}{\mathrm{kgK}}\cdot 298\,\mathrm{K}}=1{,}17\,\frac{\mathrm{kg}}{\mathrm{m}^3}.$$

And therefore the speed in the wind-tunnel can be calculated as

$$V_1 = \sqrt{\frac{2 \cdot 1000\,\frac{\mathrm{kg}}{\mathrm{m}^3} \cdot 9{,}81\,\frac{\mathrm{m}}{\mathrm{s}^2} \cdot 45\,\mathrm{mm}}{1{,}17\,\frac{\mathrm{kg}}{\mathrm{m}^3}}} 27{,}5\,\frac{\mathrm{m}}{\mathrm{s}}.$$

$$\boxed{V_1 = 27{,}5\,\frac{\mathrm{m}}{\mathrm{s}}}$$ RESULT

Problem 6.9

In a pipe 0,3 m in diameter, 0,3 $\frac{\text{m}^3}{\text{s}}$ of water are pumped up a hill. On the hilltop (elevation 48), the line reduces to 0,2 m diameter. If the pump maintains a pressure of 690 kPa at elevation 21, calculate the pressure in the pipe on the hilltop.

Assumptions

- Elevation 48 and 21 correspond to $z = 48\,\text{m}$ and $z = 21\,\text{m}$ respectively
- Steady, incompressible, inviscid, and uniform flow

Derivation

From continuity the flow rate of water must be equal at all sections of the pipe. Therefore we can employ the continuity equation as

$$Q = A_1 V_1 = A_2 V_2.$$

Hence the flow speeds at both elevations can be found as

$$\begin{aligned} V_1 &= \frac{Q}{A_1} = \frac{0{,}3\,\frac{\text{m}^3}{\text{s}}}{\pi \cdot \frac{(0{,}3\,\text{m})^2}{4}} = 4{,}24\,\frac{\text{m}}{\text{s}} \\ V_2 &= \frac{Q}{A_2} = \frac{0{,}3\,\frac{\text{m}^3}{\text{s}}}{\pi \cdot \frac{(0{,}2\,\text{m})^2}{4}} = 9{,}55\,\frac{\text{m}}{\text{s}}. \end{aligned}$$

Now we can use the Bernoulli equation

$$\begin{aligned} \frac{p_2}{\rho} + \frac{V_2^2}{2} + g z_2 &= \frac{p_1}{\rho} + \frac{V_1^2}{2} + g z_1 \\ p_2 &= p_1 + \rho \frac{V_1^2 - V_2^2}{2} + \rho g (z_1 - z_2) \\ &= 690\,\text{kPa} + 1000\,\frac{\text{kg}}{\text{m}^3} \cdot \frac{\left(4{,}24\,\frac{\text{m}}{\text{s}}\right)^2 - \left(9{,}55\,\frac{\text{m}}{\text{s}}\right)^2}{2} + 1000\,\frac{\text{kg}}{\text{m}^3} \cdot 9{,}81\,\frac{\text{m}}{\text{s}^2} \cdot (21\,\text{m} - 48\,\text{m}) \\ &= 389\,\text{kPa}. \end{aligned}$$

$$\boxed{p_2 = 389\,\text{kPa}}$$ RESULT

Problem 6.2

The velocity field for a two-dimensional downward flow of water against a plate is given by $\boldsymbol{V} = Ax\hat{\imath} - Ay\hat{\jmath}$. Plot the pressure gradient along the centerline and determine the pressure gradient at $(x, y) = (0, 2), (0, 0)$, and $(2, 0)$.

Assumptions

- Frictionless and incompressible flow

Derivation

To find the pressure gradient we use Euler's equation

$$\rho\,(\boldsymbol{V}\cdot\nabla)\,\boldsymbol{V} = \rho\boldsymbol{g} - \nabla p.$$

The particle acceleration is defined as

$$\boldsymbol{a}_p = (\boldsymbol{V}\cdot\nabla)\,\boldsymbol{V}$$

which on scalar form is

$$\begin{aligned} a_{x_p} &= u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + \frac{\partial u}{\partial t} \\ a_{y_p} &= u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} + \frac{\partial v}{\partial t}. \end{aligned}$$

Thus

$$\begin{aligned} a_{x_p} &= A^2 x \\ a_{y_p} &= A^2 y. \end{aligned}$$

Hence, the total acceleration is

$$\boldsymbol{a}_p = A^2 x + A^2 y.$$

Now we can solve the Euler equation for the pressure gradient

$$\nabla p = -\rho\boldsymbol{g} - \rho\left(A^2 x + A^2 y\right).$$

As gravity is in the y-direction this can also be written as

$$\nabla p = -\rho A^2 x\hat{\imath} - \rho\left(g + A^2 y\right)\hat{\jmath}.$$

The pressure gradient at the three points is therefore

$$\boxed{\begin{aligned} &\nabla p(0,2) = -\rho\left(g + 2A^2\right)\hat{\jmath} \\ &\nabla p(0,0) = -\rho g\hat{\jmath} \\ &\nabla p(2,0) = -2\rho A^2\hat{\imath}. \end{aligned}}$$

RESULT

Problem 6.18

Consider frictionless, incompressible flow of air over the wing of an airplane flying at 200 $\frac{\mathrm{km}}{\mathrm{h}}$. The air approaching the wing is at 65 kPa and −10 °C. At a certain point in the flow, the pressure is 60 kPa. Calculate the speed of the air relative to the wing at this point and the absolute air speed.

Assumptions

- Incompressible, inviscid, steady, and horizontal flow along a streamline

Derivation

We start by finding the density of the air at the given conditions

$$\rho = \frac{p}{RT} = \frac{65\,\mathrm{kPa}}{287\,\frac{\mathrm{J}}{\mathrm{kg\,K}} \cdot 263\,\mathrm{K}} = 0{,}861\,\frac{\mathrm{kg}}{\mathrm{m}^3}.$$

Now we can apply the Bernoulli equation as

$$\begin{aligned}
\frac{p_1}{\rho} + \frac{V_1^2}{2} &= \frac{p_2}{\rho} + \frac{V_2^2}{2} \\
V_2 &= \sqrt{V_1^2 + 2 \cdot \frac{p_1 - p_2}{\rho}} \\
&= \sqrt{\left(200\,\frac{\mathrm{km}}{\mathrm{h}}\right)^2 + 2 \cdot \frac{65\,\mathrm{kPa} - 60\,\mathrm{kPa}}{0{,}861\,\frac{\mathrm{kg}}{\mathrm{m}^3}}} \\
&= 121\,\frac{\mathrm{m}}{\mathrm{s}}.
\end{aligned}$$

This corresponds to an absolute velocity of

$$\boxed{V_{2\mathrm{abs}} = V_2 - V_1 = 65{,}4\,\frac{\mathrm{m}}{\mathrm{s}}}$$

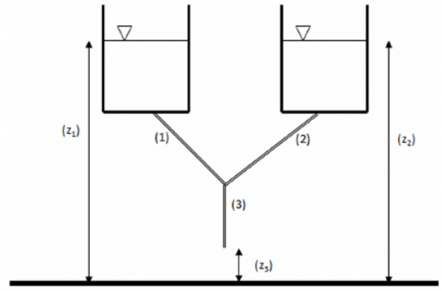
RESULT

Lecture 8: Bernoulli Equation Continued

17. September

Problem 6.23

Two water reservoirs at a 30 m elevation each have discharge pipes that are connected at a "tee" junction. One pipe has a 200 mm diameter and the other a 150 mm diameter. The outlet pipe from the "tee" is 300 mm in diameter and discharges to the atmosphere at an elevation of 20 m. Determine the total flow rate.



Assumptions

- Inviscid, incompressible, steady, immiscible, and uniform flow

Derivation

We consider a streamline from reservoir 1 and to the discharge, which we call point 4. Under the assumptions of inviscid and immiscible flow the Bernoulli equation can be applied along this streamline

$$\frac{p_1}{\rho} + \frac{V_1^2}{2} + gz_1 = \frac{p_4}{\rho} + \frac{V_4^2}{2} + gz_4.$$

In this case $p_1 = V_1 = p_4 = 0$ and therefore the outlet velocity can be found as

$$\begin{aligned} gz_1 &= \frac{V_4^2}{2} + gz_4 \\ V_4 &= \sqrt{2g\,(z_1 - z_4)} \\ &= \sqrt{2 \cdot 9{,}81\,\frac{\mathrm{m}}{\mathrm{s}^2} \cdot (30\,\mathrm{m} - 20\,\mathrm{m})} \\ &= 14{,}0\,\frac{\mathrm{m}}{\mathrm{s}}. \end{aligned}$$

Knowing the flow speed and the pipe diameter the flow rate can be found using the continuity equation as

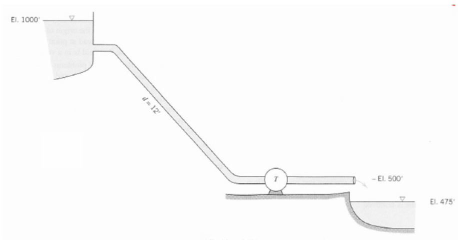
$$Q = A_4 V_4 = \pi \cdot \frac{(300\,\mathrm{mm})^2}{4} \cdot 14{,}0\,\frac{\mathrm{m}}{\mathrm{s}} = 0{,}99\,\frac{\mathrm{m}^3}{\mathrm{s}}.$$

$$\boxed{Q = 0{,}99\,\frac{\mathrm{m}^3}{\mathrm{s}}}$$

RESULT

Problem 6.28

The turbine extracts power from the water flowing from the reservoir. Find the horsepower extracted if the flow through the system is 1000 ft^3/s. Draw the energy line and the hydraulic grade line.



Assumptions

- Steady, incompressible, inviscid and uniform flow

Derivation

The flow speed through the turbine can be found from the continuity equation

$$Q = A_2 V_2 \implies V_2 = \frac{Q}{A_2} = \frac{1000\,\frac{\text{ft}^3}{\text{s}}}{\pi \cdot (6\,\text{in})^2} = 2{,}7\,\frac{\text{m}}{\text{s}}.$$

From the energy equation we get

$$\frac{p_1}{\rho} + \frac{V_1^2}{2} + g z_1 = \frac{p_2}{\rho} + \frac{V_2^2}{2} + g z_2 + T.$$

In this $p_1 = p_2 = V_1 \approx 0$, hence

$$g z_1 = \frac{V_2^2}{2} + g z_2 + T \implies T = g(z_1 - z_2) - \frac{V_2^2}{2}.$$

Dividing this by g yields the total head

$$H = z_1 - z_2 - \frac{V_2^2}{2g} = 1000\,\text{ft} - 500\,\text{ft} - \frac{\left(2{,}7\,\frac{\text{m}}{\text{s}}\right)^2}{2 \cdot 9{,}81\,\frac{\text{m}}{\text{s}^2}} = 152\,\text{m}.$$

The formula for finding the hydraulic power from a given head is

$$P_{\text{hyd}} = \rho g Q H = 1000\,\frac{\text{kg}}{\text{m}^3} \cdot 9{,}81\,\frac{\text{m}}{\text{s}^2} \cdot 1000\,\frac{\text{ft}^3}{\text{s}} \cdot 152\,\text{m} = 42{,}2\,\text{kW}.$$

$$\boxed{P_{\text{hyd}} = 42{,}2\,\text{kW}}$$ RESULT

Lecture 9: Dimensional Analysis

22. September 2025

Problem 7.10

A weir is a submerged gate extending across the width of a channel in which water is flowing. The depth of the water upstream of the weir can be used to determine the flow rate. Assume that the volume flow depends on the upstream depth, channel width, and gravity to find the non-dimensional parameters for this situation and determine the dependency of flow rate on the other variables.

Assumptions

- The flow rate depends on the upstream depth, channel width, and gravity.

Derivation

To solve this problem we will utilize the Buckingham-Pi theorem. We let Q denote the volumetric flow rate, g denote gravity, b denote the channel width and h the upstream depth, corresponding to $n = 4$ parameters. We let L and t be our fundamental dimensions and we therefore have $r = 2$ fundamental dimensions. Expressing our 4 parameters based on our two fundamental dimensions we get

$$Q \to \frac{L^3}{t}, \qquad g \to \frac{L}{t^2}, \qquad b \to L, \qquad h \to L.$$

Here we select g and h as our repeating parameters, giving us $m = 2$ repeating parameters. We will have $n - m = 2$ dimensionless groups. Setting up dimensional equations, we obtain

$$\Pi_1 = Q \cdot h^a \cdot g^b \qquad \Longrightarrow \qquad L^0 \cdot t^0 = \frac{L^3}{t} \cdot L^a \cdot \left(\frac{L}{t^2}\right)^b.$$

Now we can sum the exponents to get

$$\begin{aligned} 0 &= 3 + a + b \\ 0 &= -1 - 2b \\ \Longrightarrow b &= -\frac{1}{2} \\ \Longrightarrow a &= -\frac{5}{2}. \end{aligned}$$

Which corresponds to

$$\Pi_1 = Q \cdot h^{-\frac{5}{2}} \cdot g^{-\frac{1}{2}} = \frac{Q}{h^2 \cdot \sqrt{gh}}.$$

We can verify that this is a correct dimensionless group as

$$\left(\frac{L^3}{t}\right) \cdot \left(\frac{1}{L}\right)^2 \cdot \left(\frac{t}{L}\right) = 1.$$

For our second dimensionless group we get

$$\Pi_2 = b \cdot h^a \cdot g^b \qquad \Longrightarrow \qquad L^0 \cdot t^0 = L \cdot L^a \cdot \left(\frac{L}{t^2}\right) b.$$

Summing the exponents yields

$$\begin{aligned} 0 &= 1 + a + b \\ 0 &= -2b \\ \Longrightarrow b &= 0 \end{aligned}$$

$$\implies a = -1.$$

Which corresponds to

$$\Pi_2 = b \cdot h^{-1} \cdot g^0 = \frac{b}{h}.$$

And we can verify this as

$$L \cdot \frac{1}{L} = 1.$$

The functional relationship is $\Pi_1 = f(\Pi_2)$, or

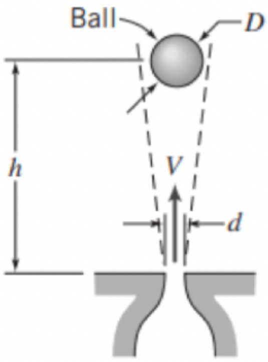
$$\frac{Q}{h^2 \cdot \sqrt{gh}} = f\left(\frac{b}{h}\right).$$

Solving for the flow rate gives

$$\boxed{Q = h^2 \sqrt{gh} \cdot f\left(\frac{b}{h}\right)}$$ RESULT

Problem 7.14

A ball is suspended in an air jet in a stable position as shown on the figure. Experiments show that the equilibrium height of the ball is found to depend on the ball diameter and weight, jet diameter and velocity, and the air density and viscosity. Determine the dimensionless parameters that characterize this experiment.



Assumptions

- The equilibrium height of the ball depends on the ball diameter and weight, the jet diameter and velocity, and the air density and viscosity.

Derivation

We will once again employ the Buckingham-Pi theorem. We let h denote the equilibrium height, d denote the ball diameter, W denote the ball weight, D denote the jet diameter V the jet velocity, ρ the air density and μ the air viscosity. This corresponds to $n = 7$ parameters. We choose mass M, length L and time t as our fundamental dimensions ($r = 3$) and get

$$h \to L, \qquad d \to L, \qquad W \to \frac{M \cdot L}{t^2}, \qquad D \to L, \qquad V \to \frac{L}{t}, \qquad \rho \to \frac{M}{L^3}, \qquad \mu \to \frac{M}{L \cdot t}.$$

We choose ρ V and d as our repeating parameters giving us $m = 3$. We will have $n - m = 4$ dimensionless groups. Setting up the first dimensional equation we obtain

$$\Pi_1 = h \cdot \rho^a \cdot V^b \cdot d^c \qquad \Longrightarrow \qquad L^0 \cdot M^0 \cdot t^0 = L \cdot \left(\frac{M}{L^3}\right)^a \cdot \left(\frac{L}{t}\right)^b \cdot L^c.$$

Summing the exponents gives

$$\begin{aligned} 0 &= 1 - 3a + b + c \\ 0 &= a \\ 0 &= -b \\ \Longrightarrow\ c &= -1. \end{aligned}$$

Which corresponds to

$$\Pi_1 = \frac{h}{d}.$$

Which can be verified as

$$L \cdot \frac{1}{L} = 1.$$

For the second dimensional equation we obtain

$$\Pi_2 = D \cdot \rho^a \cdot V^b \cdot d^c \qquad \Longrightarrow \qquad L^0 \cdot M^0 \cdot t^0 = L \cdot \left(\frac{M}{L^3}\right)^a \cdot \left(\frac{L}{t}\right)^b \cdot L^c.$$

This is the same system as before and the exponents will therefore be the same resulting in

$$\Pi_2 = \frac{D}{d}.$$

For the third dimensional equation we obtain

$$\Pi_3 = W \cdot \rho^a \cdot V^b \cdot d^c \qquad \Longrightarrow \qquad L^0 \cdot M^0 \cdot t^0 = \frac{M \cdot L}{t^2} \cdot \left(\frac{M}{L^3}\right)^a \cdot \left(\frac{L}{t}\right)^b \cdot L^c.$$

Summing exponents gives

$$\begin{aligned} 0 &= 1 - 3a + b + c \\ 0 &= 1 + a \\ 0 &= -2 - b \\ \Longrightarrow a &= -1 \\ b &= -2 \\ c &= -2. \end{aligned}$$

Which corresponds to

$$\Pi_3 = \frac{W}{\rho \cdot V^2 \cdot d^2}.$$

And the fourth dimensional equation gives

$$\Pi_4 = \mu \cdot \rho^a \cdot V^b \cdot d^c \qquad \Longrightarrow \qquad L^0 \cdot M^0 \cdot t^0 = \frac{M \cdot L}{t} \cdot \left(\frac{M}{L^3}\right)^a \cdot \left(\frac{L}{t}\right)^b \cdot L^c.$$

Summing exponents gives

$$\begin{aligned} 0 &= 1 - 3a + b + c \\ 0 &= 1 + a \\ 0 &= -1 - b \\ \Longrightarrow a &= -1 \\ b &= -1 \\ c &= -1. \end{aligned}$$

Which corresponds to

$$\Pi_4 = \frac{\mu}{\rho \cdot V \cdot d}.$$

$$\boxed{\begin{aligned} \Pi_1 &= \frac{h}{d}, \\ \Pi_2 &= \frac{D}{d}, \\ \Pi_3 &= \frac{W}{\rho \cdot V^2 \cdot d^2}, \\ \Pi_4 &= \frac{\mu}{\rho \cdot V \cdot d}. \end{aligned}}$$

RESULT

Problem 7.18

There are various applications such as freezing of food, combustion of small particles, or some chemical reactions in which a fluid flows through a matrix of particles, called a fluidized bed. The fluid pressure drop is a function of the diameter of the particles d, the flow length of the bed L, the fluid velocity V, fluid density ρ, and fluid viscosity μ. Use dimensional analysis to find the functional dependence of pressure drop on the other variables.

Assumptions

- The fluid pressure drop depends on the diameter of the particles, the flow length, the fluid velocity, the fluid density and the fluid viscosity

Derivation

We let the fluid pressure drop be termed Δp and in total we have $n = 6$ parameters. We choose mass M, length L and time t as our fundamental dimensions corresponding to $r = 3$. Our parameters expressed in terms of these are

$$\Delta p \to \frac{M}{L \cdot t^2}, \qquad d \to L, \qquad L \to L, \qquad V \to \frac{L}{t}, \qquad \rho \to \frac{M}{L^3}, \qquad \mu \to \frac{M}{L \cdot t}.$$

We choose L, V, and ρ as our repeating parameters giving $m = 3$. This requires $n - m = 3$ dimensionless groups. Setting up the first of these gives

$$\Pi_1 = \Delta p \cdot L^a \cdot V^b \cdot \rho^c \qquad \Longrightarrow \qquad L^0 M^0 t^0 = \frac{M}{L \cdot t^2} \cdot L^a \cdot \left(\frac{L}{t}\right)^b \cdot \left(\frac{M}{L^3}\right)^c.$$

Summing the exponents gives

$$\begin{aligned} 0 &= -1 + a + b - 3c \\ 0 &= 1 + c \\ 0 &= -2 - b \\ \Longrightarrow c &= -1 \\ b &= -2 \\ a &= 0. \end{aligned}$$

This corresponds to

$$\Pi_1 = \frac{\Delta p}{V^2 \rho}.$$

The second dimensionless group becomes

$$\Pi_2 = d \cdot L^a \cdot V^b \cdot \rho^c \qquad \Longrightarrow \qquad L^0 M^0 t^0 = L \cdot L^a \cdot \left(\frac{L}{t}\right)^b \cdot \left(\frac{M}{L^3}\right)^c.$$

Summing exponents gives

$$\begin{aligned} 0 &= 1 + a + b - 3c \\ 0 &= c \\ 0 &= -b \\ \Longrightarrow a &= -1. \end{aligned}$$

Which corresponds to

$$\Pi_2 = \frac{d}{L}.$$

The third dimensionless group is

$$\Pi_3 = \mu \cdot L^a \cdot V^b \cdot \rho^c \qquad \Longrightarrow \qquad L^0 M^0 t^0 = \frac{M}{L \cdot t} \cdot L^a \cdot \left(\frac{L}{t}\right)^b \cdot \left(\frac{M}{L^3}\right)^c.$$

Summing exponents gives

$$\begin{aligned}
0 &= -1 + a + b - 3c \\
0 &= 1 + c \\
0 &= -1 - b \\
\Longrightarrow b &= -1 \\
c &= -1 \\
a &= -1.
\end{aligned}$$

Which corresponds to

$$\Pi_3 = \frac{\mu}{L \cdot V \cdot \rho}.$$

The functional relationship is therefore

$$\frac{\Delta p}{\rho V^2} = f\left(\frac{d}{L}, \frac{\mu}{L \cdot V \cdot \rho}\right).$$

Here $\frac{\Delta p}{\rho V^2}$ is the pressure coefficient and $\frac{\mu}{L \cdot V \cdot \rho} = \frac{1}{\mathrm{Re}}$.

$$\boxed{\frac{\Delta p}{\rho V^2} = f\left(\frac{d}{L}, \frac{\mu}{L \cdot V \cdot \rho}\right)}$$ RESULT

Lecture 10: Internal Flows: Fully Developed Laminar Flows

24. September 2025

Problem 8.42

In the laminar flow of an oil of viscosity 1 Pa s, the velocity at the center of a 0,3 m pipe is 4,5 $\frac{\mathrm{m}}{\mathrm{s}}$ and the velocity distribution is parabolic. Calculate the shear stress at the pipe wall and within the fluid 75 mm from the pipe wall.

Assumptions

- Steady and incompressible flow

Derivation

As the velocity profile is parabolic the flow is fully developed. The general formula for the velocity distribution of a fully developed flow in a pipe is

$$u = -\frac{R^2}{4\mu}\left(\frac{\mathrm{d}p}{\mathrm{d}x}\right)\left(1-\left(\frac{r}{R}\right)^2\right).$$

In this case the diameter is $D = 0{,}3\,\mathrm{m}$ meaning the radius is $R = \frac{D}{2} = 0{,}15\,\mathrm{m}$, the flow speed us $u = 4{,}5\,\frac{\mathrm{m}}{\mathrm{s}}$ and the viscosity is $1\,\mathrm{Pa\,s}$. At the center of the pipe $r = 0$ we have

$$u = -\frac{(0{,}15\,\mathrm{m})^2}{4\cdot 1\,\mathrm{Pa\,s}}\left(\frac{\mathrm{d}p}{\mathrm{d}x}\right).$$

Here we can solve for $\frac{\mathrm{d}p}{\mathrm{d}x}$ to get

$$\begin{aligned}\frac{\mathrm{d}p}{\mathrm{d}x} &= --\frac{4{,}5\,\frac{\mathrm{m}}{\mathrm{s}}\cdot 4\cdot 1\,\mathrm{Pa\,s}}{(0{,}15\,\mathrm{m})^2}\\ &= -800\,\frac{\mathrm{Pa}}{\mathrm{m}}.\end{aligned}$$

The shear stress distribution in a pipe is

$$\tau_{rx} = \frac{r}{2}\left(\frac{\mathrm{d}p}{\mathrm{d}x}\right).$$

At the pipe wall this is

$$\tau_{\text{wall}} = \frac{R}{2}\cdot\frac{\mathrm{d}p}{\mathrm{d}x} = \frac{0{,}15\,\mathrm{m}}{2}\cdot 800\,\frac{\mathrm{Pa}}{\mathrm{m}} = 60\,\mathrm{Pa}.$$

75 mm from the pipe wall this is

$$\tau_{75\,\mathrm{mm}} = \frac{(0{,}15\,\mathrm{m} - 75\,\mathrm{mm})}{2}\cdot 800\,\frac{\mathrm{Pa}}{\mathrm{m}} = 30\,\mathrm{Pa}.$$

$$\boxed{\begin{aligned}\tau_{\text{wall}} &= 60\,\mathrm{Pa}\\ \tau_{75\,\mathrm{mm}} &= 30\,\mathrm{Pa}.\end{aligned}}$$

RESULT

Problem 8.13

When a horizontal laminar flow occurs between two parallel plates of infinite extent 0,3 m apart, the velocity at the midpoint between the plates is 2,7 $\frac{\mathrm{m}}{\mathrm{s}}$. Calculate (a) the flow rate through a section 0,9 m wide, (b) the velocity gradient at the surface of the plate, (c) the wall shearing stress if the fluid has viscosity 1,44 Pa s, (d) the pressure drop in each 30 m along the flow.

Assumptions

- Steady, incompressible, and fully-established flow

Derivation

In this case we must use the velocity distribution for flow between infinite plates,

$$u = \frac{a^2}{2\mu}\frac{\mathrm{d}p}{\mathrm{d}x}\left(\left(\frac{y}{a}\right)^2 - \frac{y}{a}\right).$$

In this particular case $a = 0{,}3\,\mathrm{m}$. For the velocity at the midpoint ($y = 0{,}15\,\mathrm{m}$) we get

$$V_c = \frac{(0{,}3\,\mathrm{m})^2}{2\cdot\mu}\frac{\mathrm{d}p}{\mathrm{d}x}\left(\left(\frac{0{,}15\,\mathrm{m}}{0{,}3\,\mathrm{m}}\right)^2 - \left(\frac{0{,}15\,\mathrm{m}}{0{,}30\,\mathrm{m}}\right)\right) = 2{,}7\,\frac{\mathrm{m}}{\mathrm{s}}.$$

And therefore the pressure gradient can be found as

$$\frac{1}{2\mu}\frac{\mathrm{d}p}{\mathrm{d}x} = \frac{2{,}7\,\frac{\mathrm{m}}{\mathrm{s}}}{(0{,}3\,\mathrm{m})^2\cdot\left(\left(\frac{0{,}15\,\mathrm{m}}{0{,}3\,\mathrm{m}}\right)^2 - \frac{0{,}15\,\mathrm{m}}{0{,}30\,\mathrm{m}}\right)} = -120\,\frac{1}{\mathrm{m\,s}}.$$

The average velocity is given by

$$\overline{V} = -\frac{1}{12\mu}\frac{\mathrm{d}p}{\mathrm{d}x}a^2.$$

Inputting known values yields

$$\overline{V} = 120\,\frac{1}{\mathrm{m\,s}}\cdot\frac{1}{6}\cdot(0{,}3\,\mathrm{m})^2 = 1{,}8\,\frac{\mathrm{m}}{\mathrm{s}}.$$

Which gives a volumetric flow rate of

$$Q = A\cdot\overline{V} = 0{,}3\,\mathrm{m}\cdot 0{,}9\,\mathrm{m}\cdot 1{,}8\,\frac{\mathrm{m}}{\mathrm{s}} = 0{,}486\,\frac{\mathrm{m}^3}{\mathrm{s}}.$$

The velocity gradient is

$$\frac{\mathrm{d}u}{\mathrm{d}y} = \frac{a^2}{2\mu}\frac{\mathrm{d}p}{\mathrm{d}x}\left(\frac{2y}{a^2} - \frac{1}{a}\right) = (0{,}3\,\mathrm{m})^2\cdot\left(-120\,\frac{1}{\mathrm{m\,s}}\right)\cdot\left(0 - \frac{1}{0{,}30\,\mathrm{m}}\right) = 36\,\frac{1}{\mathrm{s}}.$$

The shear stress at the wall is

$$\tau_{yx} = \mu\frac{\mathrm{d}u}{\mathrm{d}y} = 1{,}44\,\mathrm{Pa\,s}\cdot 36\,\frac{1}{\mathrm{s}} = 51{,}8\,\mathrm{Pa}.$$

For the pressure drop per 30 m we know that

$$\begin{aligned}\frac{1}{2\mu}\frac{\mathrm{d}p}{\mathrm{d}x} &= -120\,\frac{1}{\mathrm{m\,s}}\\ \frac{\mathrm{d}p}{\mathrm{d}x} &= -120\,\frac{1}{\mathrm{m\,s}}\cdot 2\cdot 1{,}44\,\mathrm{Pa\,s}\\ &= -346\,\frac{\mathrm{Pa}}{\mathrm{m}}\end{aligned}$$

.

Now entering the length $\Delta x = 30\,\mathrm{m}$ we get

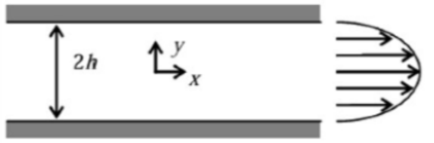
$$\Delta p_{30\,\mathrm{m}} = \frac{\mathrm{d}p}{\mathrm{d}x} \cdot (-\Delta x) = -346\,\frac{\mathrm{Pa}}{\mathrm{m}} \cdot -30\,\mathrm{m} = 10{,}4\,\mathrm{kPa}.$$

$$\boxed{\begin{aligned} Q_{0,9\,\mathrm{m}} &= 0{,}486\,\frac{\mathrm{m}^3}{\mathrm{s}} \\ \left(\frac{\mathrm{d}u}{\mathrm{d}y}\right)_{\mathrm{wall}} &= 36\,\frac{1}{\mathrm{s}} \\ \tau_{\mathrm{wall}} &= 51{,}8\,\mathrm{Pa} \\ \Delta p_{30\,\mathrm{m}} &= 10{,}4\,\mathrm{kPa}. \end{aligned}}$$

RESULT

Problem 8.6

A fully developed and laminar flow of oil occurs between parallel plates. The pressure gradient creating the flow is $1{,}25\,\frac{\text{kPa}}{\text{m}}$ of length and the channel half-width is 1,5 mm. Calculate the magnitude and length direction of the wall shear stress at the upper plate surface. Find the volume flow rate through the channel. The viscosity is $0{,}50\,\text{N s/m}^2$.



Derivation

The basic velocity distribution is

$$u = -\frac{a^2}{2\mu}\frac{\mathrm{d}p}{\mathrm{d}x}\left(1-\left(\frac{y}{a}\right)^2\right).$$

This corresponds to a velocity gradient of

$$\frac{\mathrm{d}u}{\mathrm{d}y} = -\frac{a^2}{2\mu}\frac{\mathrm{d}p}{\mathrm{d}x}\left(\frac{2y}{a^2}\right).$$

Now the shear stress distribution can be found as

$$\tau_{yx} = \mu\frac{\mathrm{d}u}{\mathrm{d}y} = -\frac{a^2}{2}\frac{\mathrm{d}p}{\mathrm{d}x}\left(-\frac{2y}{a^2}\right) = -y\frac{\mathrm{d}p}{\mathrm{d}x}.$$

At the outer edge where $y = h$ this becomes

$$\tau_{yx} = -h\frac{\mathrm{d}p}{\mathrm{d}x} = -1{,}5\,\text{mm}\cdot 1{,}25\,\frac{\text{kPa}}{\text{m}} = -1{,}88\,\text{Pa}.$$

The volume flow rate is

$$Q = \int u\,\mathrm{d}A = \int_{-h}^{h} u\cdot b\,\mathrm{d}y = -\frac{bh^2}{2\mu}\frac{\mathrm{d}p}{\mathrm{d}x}\cdot\int_{-h}^{h} 1-\left(\frac{y}{h}\right)^2\mathrm{d}y = -\frac{2h^3\cdot b}{3\mu}\frac{\mathrm{d}p}{\mathrm{d}x}.$$

Now we can solve for the flow rate per unit depth $\frac{Q}{b}$ to get

$$\frac{Q}{b} = -\frac{2}{3}\frac{h^3}{\mu}\frac{\mathrm{d}p}{\mathrm{d}x} = -\frac{2}{3}\cdot(1{,}5\,\text{mm})^3\cdot\frac{1}{0{,}50\,\frac{\text{Ns}}{\text{m}^2}}\cdot 1{,}25\,\frac{\text{kPa}}{\text{m}} = -5{,}6\,\frac{\text{mm}^2}{\text{s}}.$$

$$\boxed{\frac{Q}{b} = -5{,}6\cdot 10^{-6}\,\frac{\text{m}^2}{\text{s}}}$$

RESULT

Lecture 11: Internal Flows: Flows in Pipes and Ducts

29. September 2025

Problem 8.20

For a turbulent flow of a fluid in 0,6 m diameter pipe, the velocity 0,15 m from the wall is 2,7 $\frac{\mathrm{m}}{\mathrm{s}}$. Estimate the wall shear stress using the $1/7^{\mathrm{th}}$ expression for the velocity profile.

Assumptions

- Fully developed, steady, and incompressible turbulent flow.

Derivation

The velocity profile in the turbulent region away from a pipe wall is

$$\frac{\overline{u}}{u_*} = 2{,}5 \ln \frac{y u_*}{\nu} + 5{,}0.$$

The kinematic viscosity $\nu = 1{,}00 \cdot 10^{-6}\,\frac{\mathrm{m}^2}{\mathrm{s}}$ of water at 20 °C. Using the relation we get

$$\frac{2{,}7\,\frac{\mathrm{m}}{\mathrm{s}}}{u_*} = 2{,}5 \ln\left(\frac{0{,}15\,\mathrm{m} \cdot u_*}{1{,}00 \cdot 10^{-6}\,\frac{\mathrm{m}^2}{\mathrm{s}}}\right) + 5{,}0.$$

Solving this expression for the friction velocity u_* yields

$$u_* = 0{,}094\,\frac{\mathrm{m}}{\mathrm{s}}.$$

The relationship between the friction velocity and the wall shear stress is

$$u_* = \sqrt{\frac{\tau_w}{\rho}} \implies \tau_w = u_*^2 \cdot \rho.$$

For $\rho = 1000\,\mathrm{kg}\,\mathrm{m}^3$ we get

$$\tau_w = \left(0{,}094\,\frac{\mathrm{m}}{\mathrm{s}}\right)^2 \cdot 1000\,\frac{\mathrm{kg}}{\mathrm{m}^3} = 8{,}84\,\mathrm{Pa}.$$

$$\boxed{\tau_w = 8{,}84\,\mathrm{Pa}}$$ RESULT

Problem 8.24

A pipe runs for an elevation of 45 m to an elevation of 115 m. The inlet pressure is 8,5 MPa and the head loss is 6,9 $\frac{\text{kJ}}{\text{kg}}$. Calculate the outlet pressure for (a) the inlet at the 45 m elevation and (b) the inlet at the 115 m elevation.

Assumptions

- Incompressible and fully developed flow

Derivation

The energy equation with head loss is

$$\frac{p_1}{\rho} + \alpha_1 \frac{\overline{V}_1^2}{2} + g z_1 = \frac{p_2}{\rho} + \alpha_2 \frac{\overline{V}_2^2}{2} + g z_2 + h_{l_t}.$$

As the flow is assumed incompressible $\overline{V}_1 = \overline{V}_2$, therefore the pressure at the 115 m outlet can be found as

$$\begin{aligned} p_2 &= p_1 + \rho g (z_1 - z_2) - \rho h_{l_t} \\ &= 8{,}5\,\text{MPa} + 1000\,\frac{\text{kg}}{\text{m}^3} \cdot 9{,}81\,\frac{\text{m}}{\text{s}^2} \cdot (-70\,\text{m}) - 1000\,\frac{\text{kg}}{\text{m}^3} \cdot 6{,}9\,\frac{\text{kJ}}{\text{kg}} \\ &= 2{,}29\,\text{MPa}. \end{aligned}$$

$$\boxed{p_2 = 2{,}29\,\text{MPa}}$$ RESULT

Problem 8.28

Water flows in a smooth 0,2 m diameter pipeline that is 65 m long. The Reynolds number is 10^6. Determine the flow rate and pressure drop. After many years of use, minerals deposited on the pipe cause the same pressure drop to produce only one-half the original flow rate. Estimate the size of the relative roughness elements for the pipe.

Assumptions

- Fully developed, steady, and incompressible flow in pipe.
- Kinetic energy coefficient equal to 1.
- Constant viscosity.

Derivation

Using the definition of the Reynold's number the mean velocity in the pipe can be found as

$$\mathrm{Re} = \frac{\rho V D}{\mu} \implies V = \frac{\mathrm{Re}\cdot\mu}{\rho D} = \frac{10^6\cdot 10^{-3}\,\mathrm{Pa\,s}}{0{,}2\,\mathrm{m}\cdot 1000\,\frac{\mathrm{kg}}{\mathrm{m}^3}} = 5\,\frac{\mathrm{m}}{\mathrm{s}}.$$

This corresponds to a flow rate of

$$Q = V\cdot A = 5\,\frac{\mathrm{m}}{\mathrm{s}}\cdot\pi\cdot\frac{(0{,}20\,\mathrm{m})^2}{4} = 0{,}157\,\frac{\mathrm{m}^3}{\mathrm{s}}.$$

The pressure drop is given by

$$\frac{\Delta p}{\rho} = f\frac{L}{D}\frac{\overline{V}^2}{2} \implies \Delta p = f\frac{L}{D}\frac{\overline{V}^2\rho}{2}.$$

Here the friction factor f is defined as

$$\frac{1}{\sqrt{f}} = -2{,}0\log\left(\frac{\frac{e}{D}}{3{,}7} + \frac{2{,}51}{\mathrm{Re}\sqrt{f}}\right).$$

In this, the pipe roughness $e = 0$, hence

$$\frac{1}{\sqrt{f}} = -2{,}0\log\left(\frac{2{,}51}{\mathrm{Re}\cdot\sqrt{f}}\right) \implies f = 0{,}0116.$$

And therefore the pressure drop in the non-deteriorated pipe is,

$$\Delta p = 0{,}116\cdot\frac{65\,\mathrm{m}}{0{,}20\,\mathrm{m}}\cdot\frac{\left(5\,\frac{\mathrm{m}}{\mathrm{s}}\right)^2\cdot 1000\,\frac{\mathrm{kg}}{\mathrm{m}^3}}{2} = 47{,}1\,\mathrm{kPa}.$$

For the deteriorated pipe the head loss $H_l = \frac{\Delta p}{\rho\cdot g}$ will be the same as for the undeteriorated pipe.

$$H_L = f\frac{L}{D}\frac{\overline{V}^2}{2g}.$$

Equating the head loss in the new and the deteriorated pipes we get

$$f_{\mathrm{new}}\cdot\frac{L}{D}\cdot\frac{V_{\mathrm{new}}^2}{2g} = f_{\mathrm{det}}\cdot\frac{L}{D}\cdot\frac{\frac{V_{\mathrm{new}}^2}{4}}{2g} \implies f_{\mathrm{det}} = 4\cdot f_{\mathrm{new}} = 4\cdot 0{,}0116 = 0{,}0464.$$

Now we can user the expression for the friction factor from before and solve for $\frac{e}{D}$ to get

$$\frac{1}{\sqrt{0{,}0464}} = -2{,}0\log\left(\frac{\frac{e}{D}}{3{,}7}\frac{2{,}51}{\mathrm{Re}\cdot\sqrt{0{,}0464}}\right) \implies \frac{e}{D} = 0{,}0176.$$

$$\boxed{\begin{aligned} Q &= 0{,}157\,\frac{\mathrm{m}^3}{\mathrm{s}} \\ \Delta p &= 47{,}1\,\mathrm{kPa} \\ \frac{e}{D} &= 0{,}0176. \end{aligned}}$$

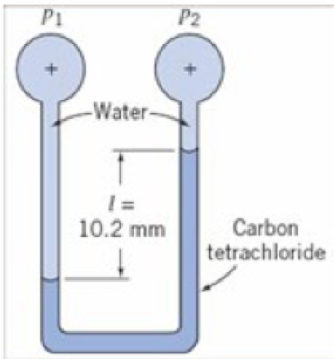
RESULT

Lecture 12: Recap

1. October 2025

Problem 3.11

Consider the two-fluid manometer shown. Calculate the applied pressure difference.



Assumptions

- Incompressible and static liquid.

Derivation

To find the pressure difference in the two-fluid manometer we will employ the hydrostatic equation with z being vertically downwards,

$$\frac{\mathrm{d}p}{\mathrm{d}z} = \rho g.$$

We let d denote the vertical distance from point 1 to the water level below point 2. The pressure contribution from the distance d at either side will cancel out and we therefore need not consider this distance. Then, moving from point 1 to point 2 using the hydrostatic equation yields

$$p_1 + \rho_\mathrm{w} \cdot g \cdot l - \rho_\mathrm{ct} \cdot g \cdot l = p_2.$$

Solving for the pressure difference $\Delta p = p_2 - p_1$ we get

$$\Delta p = \rho_\mathrm{w} \cdot g \cdot l - \rho_\mathrm{ct} \cdot g \cdot l = \left(\rho_\mathrm{w} - \rho_\mathrm{ct}\right) g l.$$

Inputting the different quantities yields

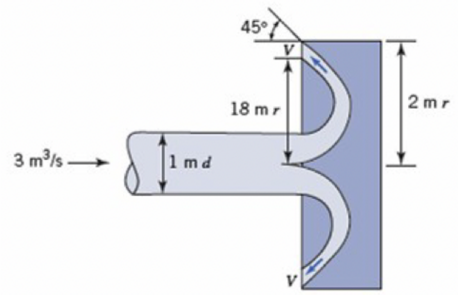
$$\Delta p = \left(1000\,\frac{\mathrm{kg}}{\mathrm{m}^3} - 1590\,\frac{\mathrm{kg}}{\mathrm{m}^3}\right) \cdot 9{,}81\,\frac{\mathrm{m}}{\mathrm{s}^2} \cdot 10{,}2 \cdot 10^{-3}\,\mathrm{m} = -59{,}1\,\mathrm{Pa}.$$

Therefore the pressure at point 2 will be 59,1 Pa lower than that at point 1.

$$\boxed{\Delta p = -59{,}1\,\mathrm{Pa}}$$ RESULT

Problem 4.10

Find V for this mushroom cap on a pipeline.



Assumptions

- Steady and incompressible flow

Derivation

The continuity equation for steady flow is

$$0 = \int_{\mathrm{CS}} \rho \boldsymbol{V} \cdot \mathrm{d}\boldsymbol{A}.$$

Since the density is constant the flow rate $Q = V \cdot A$ is constant. Hence,

$$Q_{\mathrm{in}} = Q_{\mathrm{out}}.$$

The flow rate out of the cap can also be written as

$$Q_{\mathrm{out}} = V \cdot \cos 45^\circ \cdot A_{\mathrm{out}}.$$

Combining these yields

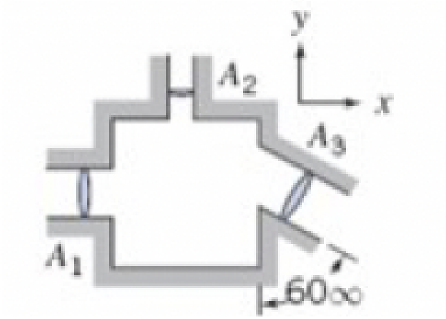
$$\begin{aligned}
Q_{\mathrm{in}} &= V \cdot \cos 45^\circ \cdot A_{\mathrm{out}} \\
V &= \frac{Q_{\mathrm{in}}}{\cos 45^\circ \cdot A_{\mathrm{out}}} \\
&= \frac{3\,\frac{\mathrm{m}^3}{\mathrm{s}}}{\cos 45^\circ \cdot \pi \cdot \left((2\,\mathrm{m})^2 - (1{,}8\,\mathrm{m})^2\right)} \\
&= 1{,}78\,\frac{\mathrm{m}}{\mathrm{s}}.
\end{aligned}$$

$$\boxed{V = 1{,}78\,\frac{\mathrm{m}}{\mathrm{s}}}$$

RESULT

Problem 4.4

Fluid with 1040 $\frac{\mathrm{kg}}{\mathrm{m}^3}$ density is flowing steadily through the rectangular box shown. Given $A_1 = 0{,}046\,\mathrm{m}^2$, $A_2 = 0{,}009\,\mathrm{m}^2$, $A_3 = 0{,}056\,\mathrm{m}^2$, $\boldsymbol{V}_1 = \left(3\,\frac{\mathrm{m}}{\mathrm{s}}\right)\hat{\imath}$, and $\boldsymbol{V}_2 = \left(6\,\frac{\mathrm{m}}{\mathrm{s}}\right)\hat{\jmath}$. Determine the velocity $\boldsymbol{V}_3$.



Assumptions

- Incompressible, steady and uniform flow.

Derivation

Due to the assumptions of uniform and incompressible steady flow the continuity equation reduces to

$$\sum_{CS} \boldsymbol{V} \cdot \boldsymbol{A} = 0.$$

Using the convention of $\boldsymbol{A}$ being outwards we get

$$\begin{aligned} V_1 \cdot A_1 + V_2 \cdot A_2 + V_3 \cdot A_3 &= 0 \\ V_3 &= -\frac{V_1 \cdot A_1 + V_2 \cdot A_2}{A_3} \\ &= \frac{3\,\frac{\mathrm{m}}{\mathrm{s}} \cdot 0{,}046\,\mathrm{m}^2 - 6\,\frac{\mathrm{m}}{\mathrm{s}} \cdot 0{,}009\,\mathrm{m}^2}{0{,}056\,\mathrm{m}^2} \\ &= 1{,}5\,\frac{\mathrm{m}}{\mathrm{s}}. \end{aligned}$$

Using trigonometry this can be written on Cartesian form as

$$\begin{aligned} V_{3x} &= V_3 \cdot \cos -30° = 1{,}5\,\frac{\mathrm{m}}{\mathrm{s}} \cdot \cos -30° = 1{,}3\,\frac{\mathrm{m}}{\mathrm{s}} \\ V_{3y} &= V_3 \cdot \sin -30° = 1{,}5\,\frac{\mathrm{m}}{\mathrm{s}} \cdot \sin -30° = -0{,}75\,\frac{\mathrm{m}}{\mathrm{s}} \\ \boldsymbol{V}_3 &= V_{3x}\hat{\imath} + V_{3y}\hat{\jmath} = 1{,}3\,\frac{\mathrm{m}}{\mathrm{s}}\,\hat{\imath} - 0{,}75\,\frac{\mathrm{m}}{\mathrm{s}}\,\hat{\jmath}. \end{aligned}$$

$$\boxed{\boldsymbol{V}_3 = 1{,}3\,\frac{\mathrm{m}}{\mathrm{s}}\,\hat{\imath} - 0{,}75\,\frac{\mathrm{m}}{\mathrm{s}}\,\hat{\jmath}}$$ RESULT

Problem 5.19

A velocity field is given by $\boldsymbol{V} = 2\hat{\imath} - 4x\hat{\jmath}\,\frac{\mathrm{m}}{\mathrm{s}}$. Determine an equation for the streamline. Calculate the vorticity of the flow.

Assumptions

- The flow is steady and incompressible

Derivation

The stream function for a steady velocity field (u, v) is defined as

$$u \equiv \frac{\partial \psi}{\partial y} \qquad v \equiv -\frac{\partial \psi}{\partial x}.$$

Using $u = 2\,\frac{\mathrm{m}}{\mathrm{s}}$ we get

$$\frac{\partial \psi}{\partial y} = 2.$$

By integration we obtain

$$\psi = 2y + f(x).$$

Using $v = -4x$ we get

$$-\frac{\partial \psi}{\partial x} = -4x \implies \frac{\partial \psi}{\partial x} = 4x.$$

By integration this becomes

$$\psi = 2x^2 + f(y).$$

Hence, the stream function is:

$$\psi = 2x^2 + 2y + c$$

where c is a constant. The vorticity ξ is given by

$$\begin{aligned} \xi &\equiv \nabla \times \boldsymbol{V} \\ &= \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right)\hat{k} \\ &= (-4 - 0)\,\hat{k} \\ &= -4\,\hat{k}. \end{aligned}$$

$$\boxed{\begin{gathered} \psi(x, y) = 2x^2 + 2y + c \\ \boldsymbol{\xi} = -4\,\hat{k} \end{gathered}}$$

RESULT

Lecture 16: Boundary Layer Theory

29. October 2025

Problem 9.2

For a flow at 9,0 $\frac{\text{m}}{\text{s}}$ over a smooth flat plate, estimate the maximum length of the laminar layer for air and for water as the fluid.

Assumptions

- The transition to turbulent happens at a length Reynolds number of exactly $5 \cdot 10^5$
- Uniform and incompressible flow at 20 °C

Derivation

The length Reynolds number is defined as

$$\text{Re}_x = \frac{u \cdot x}{\nu} \implies x = \frac{\text{Re}_x \cdot \nu}{u}.$$

The kinematic viscosity of air at 20 °C is approximately $\nu_{\text{air}} = 1{,}5 \cdot 10^{-5}\,\frac{\text{m}^2}{\text{s}}$ (Fig. A.3. in Appendix A of the course literature). Using this in the above equation for the distance before the flow turns turbulent for the flow of air under the given assumptions and conditions, we get:

$$x_{\text{air}} = \frac{5 \cdot 10^5 \cdot 1{,}5 \cdot 10^{-5}\,\frac{\text{m}^2}{\text{s}}}{9{,}0\,\frac{\text{m}}{\text{s}}} = 0{,}83\,\text{m}.$$

From the same figure as the kinematic viscosity of the air was found the kinematic viscosity of water at 20 °C is determined to be approximately $9 \cdot 10^{-7}\,\frac{\text{m}^2}{\text{s}}$. Thus, for water we get a turbulent transition after approximately:

$$x_{\text{water}} = \frac{5 \cdot 10^5 \cdot 9 \cdot 10^{-7}\,\frac{\text{m}^2}{\text{s}}}{9{,}0\,\frac{\text{m}}{\text{s}}} = 0{,}05\,\text{m}.$$

$$\boxed{\begin{aligned} x_{\text{air}} &= 0{,}83\,\text{m} \\ x_{\text{water}} &= 0{,}05\,\text{m}. \end{aligned}}$$

RESULT

Problem 9.8

Air at 5 $\frac{\text{m}}{\text{s}}$, atmospheric pressure, and 20 °C flows over both sides of a flat plate that is 0,8 m long and 0,3 m wide. Determine the total drag force on the plate. Determine the total drag force when the single plate is replaced by two plates each 0,4 m long and 0,3 m wide. Explain why there is a difference in the total drag even though the total surface area is the same.

Assumptions

- Steady and incompressible flow
- The plate is infinitesimally thin

Derivation

The drag force is given by

$$F_D = \frac{1}{2}\rho V^2 C_D A.$$

For air at 20 °C the density can, in Table A.10 in appendix A of the course literature be found to be approximately $\rho_{\text{air}} = 1{,}21\,\frac{\text{kg}}{\text{m}^3}$. To find the drag coefficient we must use the Reynolds number. This is given as

$$\text{Re} = \frac{VL}{\nu}.$$

For the case with the single plate we thus get

$$\text{Re} = \frac{5\,\frac{\text{m}}{\text{s}} \cdot 0{,}8\,\text{m}}{1{,}5 \cdot 10^{-5}\,\frac{\text{m}^2}{\text{s}}} = 2{,}67 \cdot 10^5.$$

The drag coefficient for flow over a flat plate is

$$C_D = \frac{1{,}33}{\sqrt{\text{Re}_L}} = \frac{1{,}33}{\sqrt{2{,}67 \cdot 10^5}} = 2{,}57 \cdot 10^{-3}.$$

Now, plugging these values into the formula for the drag force (remembering that both sides experience drag) we get:

$$(F_D)_{1\text{ plate}} = \frac{1}{2} \cdot 1{,}21\,\frac{\text{kg}}{\text{m}^3} \cdot \left(5\,\frac{\text{m}}{\text{s}}\right)^2 \cdot 2{,}57 \cdot 10^{-3} \cdot 0{,}8\,\text{m} \cdot 0{,}3\,\text{m} \cdot 2 = 1{,}87 \cdot 10^{-2}\,\text{N}.$$

For the case of two plates we have

$$\text{Re} = \frac{5\,\frac{\text{m}}{\text{s}} \cdot 0{,}4\,\text{m}}{1{,}5 \cdot 10^{-5}\,\frac{\text{m}^2}{\text{s}}} = 1{,}33 \cdot 10^5$$

$$C_D = \frac{1{,}33}{\sqrt{1{,}33 \cdot 10^5}} = 3{,}64 \cdot 10^{-3}$$

$$(F_D)_{2\text{ plates}} = \frac{1}{2} \cdot 1{,}21\,\frac{\text{kg}}{\text{m}^3} \cdot \left(5\,\frac{\text{m}}{\text{s}}\right)^2 \cdot 3{,}64 \cdot 10^{-3} \cdot 2 \cdot 2 \cdot 0{,}4\,\text{m} \cdot 0{,}3\,\text{m} = 2{,}64 \cdot 10^{-2}\,\text{N}.$$

The difference in the drag here is due to the difference between the Reynolds numbers for the two flows. This difference gives rise to a difference between the boundary layers.

$$\boxed{\begin{aligned}(F_D)_{1\text{ plate}} &= 1{,}87 \cdot 10^{-2}\,\text{N}\\ (F_D)_{2\text{ plates}} &= 2{,}64 \cdot 10^{-2}\,\text{N}.\end{aligned}}$$

RESULT

Problem 9.7

For the flow of air over a flat plate, plot on the same graph the laminar boundary-layer thickness as a function of distance along the plate, up to the point of transition, for freestream speeds of $1\,\frac{\mathrm{m}}{\mathrm{s}}$, $3\,\frac{\mathrm{m}}{\mathrm{s}}$ and $5\,\frac{\mathrm{m}}{\mathrm{s}}$. Draw some conclusions from your plot.

Assumptions

- The critical Reynolds number is $5 \cdot 10^5$
- The flow is happening at $20\,°\mathrm{C}$

Derivation

For laminar flow over a plate, the thickness of the boundary layer δ changes with length x as

$$\frac{\delta}{x} = \frac{5{,}48}{\sqrt{\mathrm{Re}_x}} = 5{,}48\sqrt{\frac{\nu x}{U}}.$$

Solving for the critical length in the Reynolds number formula we obtain

$$\begin{aligned} x &= 5 \cdot 10^5 \frac{\nu}{U} \\ x_{1\,\frac{\mathrm{m}}{\mathrm{s}}} &= 5 \cdot 10^5 \cdot \frac{1{,}5 \cdot 10^{-5}\,\frac{\mathrm{m}^2}{\mathrm{s}}}{1\,\frac{\mathrm{m}}{\mathrm{s}}} = 7{,}5\,\mathrm{m} \\ x_{3\,\frac{\mathrm{m}}{\mathrm{s}}} &= 5 \cdot 10^5 \cdot \frac{1{,}5 \cdot 10^{-5}\,\frac{\mathrm{m}^2}{\mathrm{s}}}{3\,\frac{\mathrm{m}}{\mathrm{s}}} = 2{,}5\,\mathrm{m} \\ x_{5\,\frac{\mathrm{m}}{\mathrm{s}}} &= 5 \cdot 10^5 \cdot \frac{1{,}5 \cdot 10^{-5}\,\frac{\mathrm{m}^2}{\mathrm{s}}}{5\,\frac{\mathrm{m}}{\mathrm{s}}} = 1{,}5\,\mathrm{m}. \end{aligned}$$

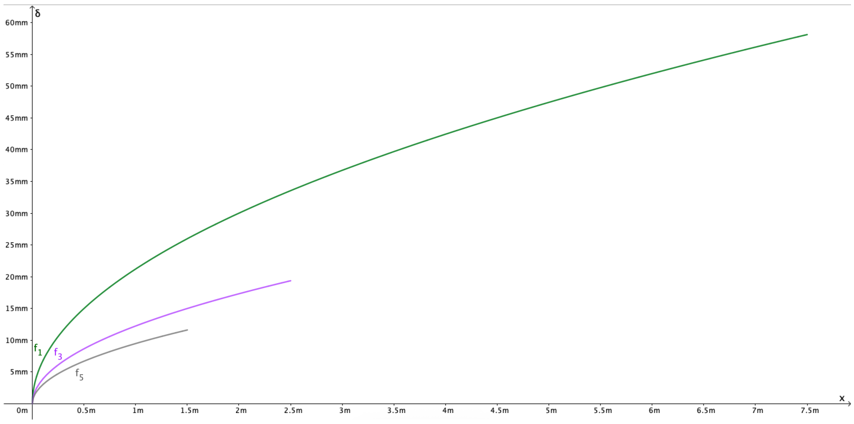


Figure 0.6: Plot of boundary layer thickness δ as a function of distance x.

$$\boxed{\begin{aligned} x_{1\,\frac{\mathrm{m}}{\mathrm{s}}} &= 5 \cdot 10^5 \cdot \frac{1{,}5 \cdot 10^{-5}\,\frac{\mathrm{m}^2}{\mathrm{s}}}{1\,\frac{\mathrm{m}}{\mathrm{s}}} = 7{,}5\,\mathrm{m} \\ x_{3\,\frac{\mathrm{m}}{\mathrm{s}}} &= 5 \cdot 10^5 \cdot \frac{1{,}5 \cdot 10^{-5}\,\frac{\mathrm{m}^2}{\mathrm{s}}}{3\,\frac{\mathrm{m}}{\mathrm{s}}} = 2{,}5\,\mathrm{m} \\ x_{5\,\frac{\mathrm{m}}{\mathrm{s}}} &= 5 \cdot 10^5 \cdot \frac{1{,}5 \cdot 10^{-5}\,\frac{\mathrm{m}^2}{\mathrm{s}}}{5\,\frac{\mathrm{m}}{\mathrm{s}}} = 1{,}5\,\mathrm{m}. \end{aligned}}$$

RESULT

Lecture 17: Fluid Flows About Immersed Bodies – Drag, Lift and Airfoils 3. November 2025

Problem 9.28

The lift and drag coefficients for a rectangular wing with a 10 m chord at takeoff are 1,0 and 0,05, respectively. Determine the span needed to lift 3560 kN at a take-off speed of 282 $\frac{\text{km}}{\text{h}}$. Determine the wing drag at this condition.

Problem 9.30

The mean velocity over the top of a wing with a 1,8 m chord moving through air at 33,5 $\frac{\text{m}}{\text{s}}$ is 40 $\frac{\text{m}}{\text{s}}$, and that over the bottom of the wing is 31 $\frac{\text{m}}{\text{s}}$. Determine the lift per meter of span and the lift coefficient.

Problem 9.6RP

A high voltage transmission line cable 10 cm in diameter runs between two towers 500 m apart. Determine the force on the cable for a wind speed of 15 $\frac{\text{m}}{\text{s}}$. Determine whether the total force would be less of four 5 cm cables with the same electrical capacity were used.
